WORKED SOLUTION

STEP III 2019, Question 2

Pure • Functional Equations

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 3.8, *Functions from Their Addition Laws*: https://beyondstep.keithmcnulty.org/ch03/addition-laws.html

The key idea

A functional equation tells us everything about a function once we know how it behaves near one point. Putting $x = y = 0$ pins down the value at 0; then writing the difference quotient at a general x and using the functional equation turns it into the difference quotient *at* 0, where we know the limit is k. That gives a differential equation, which we solve with the initial value we found.

Solution

Part (i)

The value at 0**.** Put $x = y = 0$ in the functional equation: $\mathrm{f}(0) = \mathrm{f}(0)^2$, so $\mathrm{f}(0) = 0$ or $\mathrm{f}(0) = 1$.

Suppose $\mathrm{f}(0) = 0$. Then putting $y = 0$ gives $\mathrm{f}(x) = \mathrm{f}(x)\mathrm{f}(0) = 0$ for every x, so f is the zero function and $\mathrm{f}'(0) = 0$. This contradicts $\mathrm{f}'(0) = k \neq 0$. Hence $\mathrm{f}(0) = 1$.

The derivative. From $(*)$ and the functional equation with $y = h$,

$$\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x)\mathrm{f}(h) - \mathrm{f}(x)}{h} = \mathrm{f}(x) \lim_{h\to 0} \frac{\mathrm{f}(h) - 1}{h}.$$

Since $\mathrm{f}(0) = 1$, the last limit is $\lim_{h\to 0} \dfrac{\mathrm{f}(0+h) - \mathrm{f}(0)}{h}$, which is $\mathrm{f}'(0) = k$ by $(*)$ at $x = 0$. So

$$\mathrm{f}'(x) = k\,\mathrm{f}(x),$$

as required. (We could take $\mathrm{f}(x)$ outside the limit because it does not depend on h.)

Solving. Rather than divide by f (which needs $f \neq 0$), consider $u(x) = e^{-kx} f(x)$. Then

$$u'(x) = e^{-kx}\big(f'(x) - k\,f(x)\big) = 0,$$

so u is constant, equal to $u(0) = f(0) = 1$. Therefore

$$f(x) = e^{kx}.$$

This does satisfy everything: $e^{k(x+y)} = e^{kx}e^{ky}$ and its derivative at 0 is k.

Part (ii)

The value at 0. Put $x = y = 0$:

$$g(0) = \frac{2g(0)}{1+g(0)^2} \quad \Longrightarrow \quad g(0)\big(g(0)^2 - 1\big) = 0,$$

so $g(0) \in \{0, 1, -1\}$. Since $|g(x)| < 1$ for all x, we must have $g(0) = 0$. Notice also that $|g| < 1$ guarantees $1 + g(x)g(y) > 0$, so the functional equation never divides by zero.

The difference quotient. Using the functional equation with $y = h$ and putting everything over one denominator,

$$g(x+h) - g(x) = \frac{g(x) + g(h) - g(x) - g(x)^2 g(h)}{1 + g(x)g(h)} = \frac{g(h)\big(1 - g(x)^2\big)}{1 + g(x)g(h)}.$$

Dividing by h,

$$\frac{g(x+h) - g(x)}{h} = \big(1 - g(x)^2\big) \cdot \frac{g(h)}{h} \cdot \frac{1}{1 + g(x)g(h)}.$$

Ask yourself What happens to each factor on the right as $h \to 0$?

The first factor does not involve h. Since $g(0) = 0$, the middle factor is $\dfrac{g(0+h) - g(0)}{h}$, which tends to $g'(0) = k$. For the last factor we need $g(h) \to g(0) = 0$: this holds because g is differentiable, hence continuous, at 0. So the last factor tends to 1, and

$$g'(x) = k\big(1 - g(x)^2\big).$$

Solving. Because $|g(x)| < 1$, the factor $1 - g(x)^2$ is never zero, so we may separate the variables:

$$\int \frac{dg}{1 - g^2} = \int k\,dx.$$

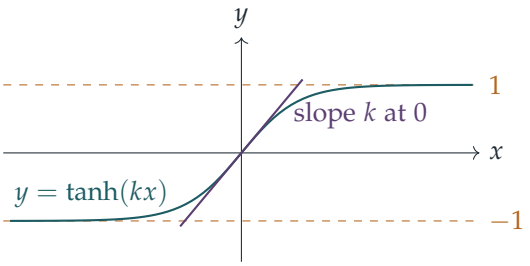
By partial fractions, $\dfrac{1}{1-g^2} = \dfrac{1}{2}\Big(\dfrac{1}{1+g} + \dfrac{1}{1-g}\Big)$, and both $1 \pm g$ are positive, so

$$\frac{1}{2}\ln\frac{1+g(x)}{1-g(x)} = kx + c.$$

At $x = 0$ the left-hand side is $\frac{1}{2}\ln 1 = 0$, so $c = 0$. Hence $\dfrac{1+g}{1-g} = e^{2kx}$, and solving for g,

$$g(x) = \frac{e^{2kx} - 1}{e^{2kx} + 1} = \tanh(kx).$$

This indeed works: $|\tanh(kx)| < 1$, the addition formula $\tanh(a+b) = \dfrac{\tanh a + \tanh b}{1 + \tanh a \tanh b}$ is exactly the functional equation, and the derivative of $\tanh(kx)$ at 0 is k.



Checks

- In (i), $f(x) = e^{kx}$ with $k = 1$, $x = 0.3$, $y = 0.5$: $e^{0.3}e^{0.5} = 1.3499 \times 1.6487 = 2.2255 = e^{0.8}$.
- In (ii), with $k = 1$, $x = 0.3$, $y = 0.5$: $\tanh 0.3 = 0.29131$, $\tanh 0.5 = 0.46212$, and $(0.29131 + 0.46212)/(1 + 0.29131 \times 0.46212) = 0.66404 = \tanh 0.8$.
- Both answers were also confirmed by computer for many random k, x, y, including the derivative relations $f' = kf$ and $g' = k(1 - g^2)$.
- The two parts are linked: $\tanh(kx) = \dfrac{e^{2kx} - 1}{e^{2kx} + 1}$ is built from the exponential of part (i), with $2k$ in place of k.

Common slips

- Discarding $f(0) = 0$ or $g(0) = \pm 1$ without a reason. The conditions $k \neq 0$ and $|g| < 1$ are there precisely to rule these out.
- Letting $g(h) \to 0$ inside the denominator without saying why: it needs continuity of g at 0, which comes from differentiability.
- Dividing by f or $1 - g^2$ without checking it is non-zero.

Beyond STEP

These are examples of *Cauchy-type functional equations*. A first analysis course shows that the exponential equation $f(x + y) = f(x)f(y)$ forces $f(x) = e^{kx}$ (or $f \equiv 0$) under

assumptions much weaker than differentiability, such as continuity at a single point, while without any regularity there are wild solutions. The tanh equation is the addition law for velocities in special relativity, where $\tanh^{-1}$ of the velocity (the rapidity) simply adds. For the analysis behind this, see Abbott, *Understanding Analysis*.

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